Backup-configuration.ps1
User / Installation guide

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Writer: Thomas Burger
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This document provides information about the usage and the work of “backup-configuration.ps1”.

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General information / purpose

This script was built to provide a convenient automated backup of SSY-Configuration on each node.

Configuration-Backups are needed in DR-Scenarios as well as hardware-replacements. Since many customers do not create them on a regular basis they may find themselves in the situation of “need but don´t have”.

Terms and their usage

Server-Group CIFS Shares: The shares used (and somehow hard-coded) for syncing backups within the SSY servergroup.

Remote Share: A windows share on another server independent from the server group. May be an archive- or backup-system.

TTL: The “Time To Live” of backups. After the amount of days these backups will get deleted.

Governor: The governor is elected within the server group and ensures that the script only runs once on each trigger. This ensures consistency and is necessary to avoid conflicts in actions (multiple concurrent configuration backups for instance). Servers in remote server groups are not part of the election process.

Requirements

Important: The script uses Powershell 4.0 and SSY commandlets. It was developed and tested for SSY 10.0 onwards and Server 2012R2. Other versions of SSY and Windows may work but are not tested.

Important: The script only provides backup to disk functionality. It is highly recommended to fetch the backups through a backup server and store it to tape. Use the “remote-share” option to avoid installing unnecessary backup-clients on the SSY servers.

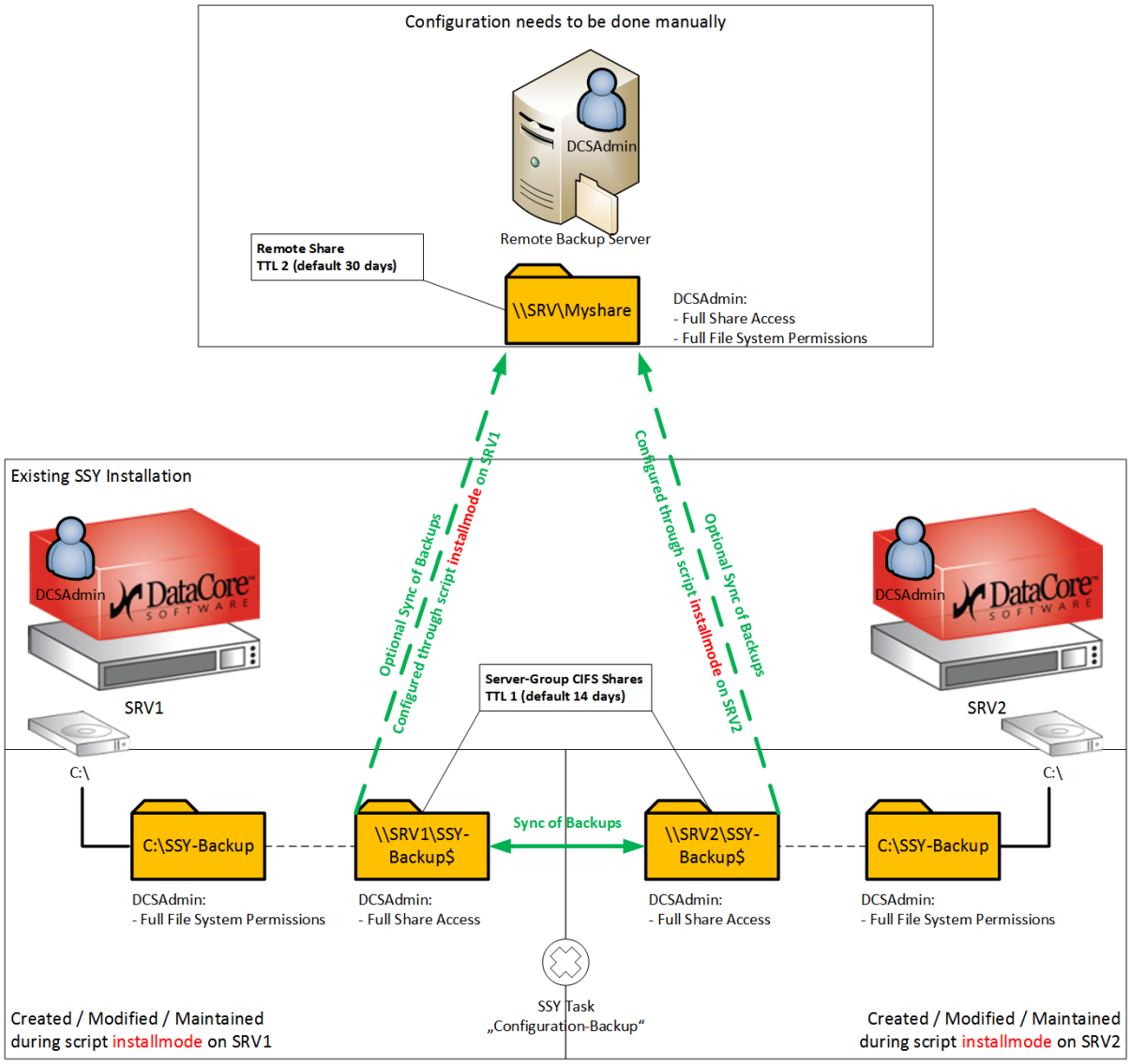
Important: If you plan to use a remote backup UNC path you can´t use administrative shares like c$ or d$. You have to provide a share name and grant access to DCSAdmin (Full) to the share as well as to the underlying NTFS folder!

Important: Syncing of the configuration backups uses Windows integrated authentication and CIFS. Please make sure the systems can communicate on all necessary ports.

Important: Do not modify the task created through “installation” of the script except for changing the triggers. Reissue the script in installation mode with modified parameters to update the configuration of the script.

Architecture Overview

The following picture describes the architecture of the backup-configuration script.



Installation / Implementation

Important: The script needs to be "installed" on each node in the SSY server group. This is necessary because it sets up the necessary environment (Server Group CIFS Shares) as well as the SSY task, trigger and action. This means that the script needs to be run on each new server joining the server group in >**installation mode (-installScript $true)**<. It is also necessary that you use the same parameters during install as this information is stored with the local server. Same paths need to be provided on each instance of the script otherwise it will fail!

The install-mode of the script can be re-run anytime to modify the settings so if you are not sure how to use the functionality just leave the settings on default. Also refer to the FAQ at the end of the document.

Default settings are:

Parameter	Usage	Value
ssyvServerBackupFolder	Local Directory of SSY Backups	C:\SSY-Backup\
ssyvServerBackupTTL	Days to store backups within the servergroup	14
additionalUNCPathFolder	The UNC path to additional CIFS share	(empty string)
additionalUNCBackupTTL	Days to store backups on the remote backup CIFS share	30

Installation

Choose a directory (recommendation something similar to *C:\Scripts\SSY-Backup*) and copy the script as well as the provided SHA512 file into that directory.

The first node within the group

Run the script (default parameters used) for installation on the **first node in the server group**:

```
PS C:\Scripts\SSY-V-Backup> .\backup-configuration.ps1 -installScript $true_
```

The script should provide an output like this:

```
2015-03-23__16-45-54  |  INFORMATION  |  create-powershellscripttask : Checking if the task exists.
2015-03-23__16-45-54  |  INFORMATION  |  create-powershellscripttask : task >Configuration-Backup< does not exist in the
 servergroup.
2015-03-23__16-45-54  |  INFORMATION  |  create-powershellscripttask : creating the task >Configuration-Backup<.
2015-03-23__16-45-54  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-45-54  |  INFORMATION  |  create-powershellscripttask : Checking for trigger in task.
2015-03-23__16-45-56  |  INFORMATION  |  create-powershellscripttask : no trigger found in task.
2015-03-23__16-45-56  |  INFORMATION  |  create-powershellscripttask : creating scheduled trigger for each day on 18:00
(equals 06:00 PM).
2015-03-23__16-45-56  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-45-56  |  INFORMATION  |  create-powershellscripttask : Checking if there is an action for this server >A
ppliance-1<.
2015-03-23__16-45-58  |  INFORMATION  |  create-powershellscripttask : no action found in task for this server.
2015-03-23__16-45-58  |  INFORMATION  |  create-powershellscripttask : creating action to perform backup script on this
server.
2015-03-23__16-45-58  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-45-58  |  VERBOSE      |  dcsService-Connection : Function invoked with parameters: >privuser< is >&lt;, >pr
ivPassword< is >**hidden**<, >privHostname< is >&lt;, >privAction< is >disconnect<, >privDataCoreServersession< is >Applian
ce-1<, >privUsePassthroughAuth< is >&lt;, >privPowershellversion< is >&lt;.
2015-03-23__16-45-58  |  INFORMATION  |  dcsService-Connection : Disconnecting from DataCore Server Session.
2015-03-23__16-45-58  |  SUCCESS      |  dcsService-Connection : success.
2015-03-23__16-45-58  |  SUCCESS      |  create-powershellscripttask : returns >true<
2015-03-23__16-45-58  |  WARNING      |  stage-2 : exiting script due to installation mode.
```

Notice the following:

1. Script creates the local backup-folder (in this case C:\SSY-Backup):

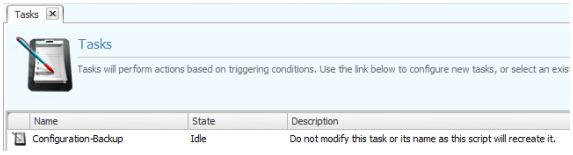
   This PC ▸ System (C:) ▸ SSY-V-Backup

2. Script also creates an CIFS share based on the folder-name

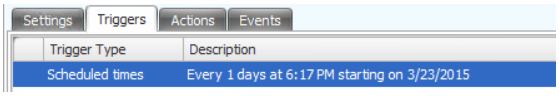
```
PS C:\Scripts\SSY-V-Backup> get-smbshare SSY-V*

Name                        ScopeName                   Path
----                        ---------                   ----
SSY-V-Backup$               *                           C:\SSY-V-Backup
```

If you provide another path, let´s say “C:\This\is\important” then the script creates a share with the name “important$”. DCSAdmin as well as local Administrator will have full access to that share.

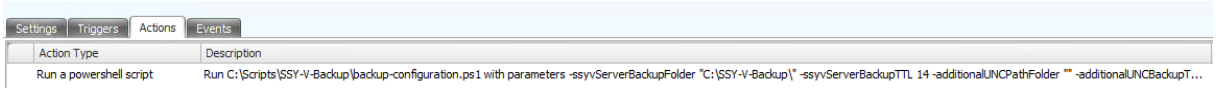
3. In the SSY GUI there will be a new task. Do not modify the name of this task!



4. Within the task there will be a new trigger. You may delete this trigger and create new ones. The script will only check (during install-mode) if there is a trigger and if not re-creates it. Default start time is on each day at 18:17 or 06:17 PM.



5. In actions you will find a new action which refers to the SSY server you recently “installed”



Important: Do not run the task yet. If you do so it will fail because the other servers in the group are not able to receive syncs yet (CIFS share not set up yet).

On each additional server in the group

The installation is exactly the same. Run the script (default parameters used) for installation on the **all other servers in the group**:

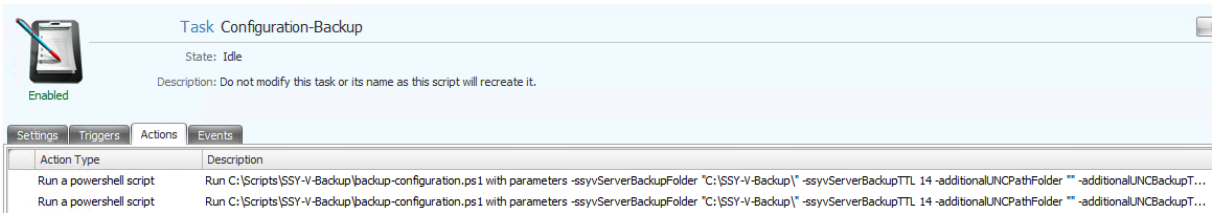
```
PS C:\Scripts\SSY-V-Backup> .\backup-configuration.ps1 -installScript $true_
```

The script will provide a slightly different output as the task and trigger is already there:

```
2015-03-23__16-43-23  |  SUCCESS      |  dcsService-Connection : success.
2015-03-23__16-43-23  |  INFORMATION  |  create-powershellscripttask : Checking if the task exists.
2015-03-23__16-43-23  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-43-23  |  INFORMATION  |  create-powershellscripttask : Checking for trigger in task.
2015-03-23__16-43-25  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-43-25  |  INFORMATION  |  create-powershellscripttask : Checking if there is an action for this server >A
ppliance-1<.
2015-03-23__16-43-27  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__16-43-27  |  VERBOSE      |  dcsService-Connection : Function invoked with parameters: >privuser< is >&lt;, >pr
ivPassword< is >**hidden**<, >privHostname< is >&lt;, >privAction< is >disconnect<, >privDataCoreServersession< is >Applian
ce-1<, >privUsePassthroughAuth< is >&lt;, >privPowershellversion< is >&lt;.
2015-03-23__16-43-27  |  INFORMATION  |  dcsService-Connection : Disconnecting from DataCore Server Session.
2015-03-23__16-43-27  |  SUCCESS      |  dcsService-Connection : success.
2015-03-23__16-43-27  |  SUCCESS      |  create-powershellscripttask : returns >true<
2015-03-23__16-43-27  |  WARNING      |  stage-2 : exiting script due to installation mode.
```

You will see exactly the same behavior in regards of folders and shares as on the first node.

In the SSY GUI you will see that in the task there will be another action created for the new server:



Troubleshooting installation

In the case installation goes wrong please check the log files provided in the script folder:

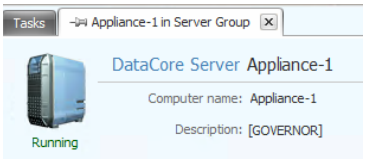
Name	Date modified	T
log	3/23/2015 3:57 PM	F
backup-configuration.ps1	3/23/2015 4:15 PM	W
backup-configuration.ps1.sha512	3/23/2015 4:16 PM	S

For each time the script is run a log file is created.

Technical overview steps executed by the script

The script works with these steps:

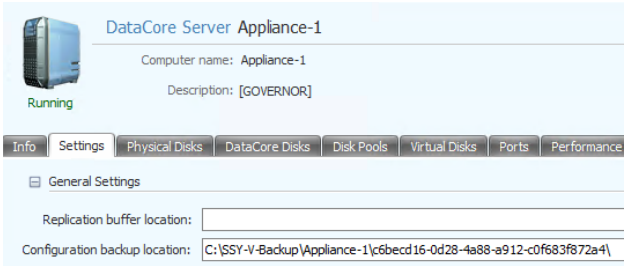
1. Script starts and creates logging environment
2. Script checks if the local server is the [GOVERNOR] and elects a governor if necessary. The governor is responsible for carrying out the backup. You can check which server is the governor by inspecting the Description of the server:

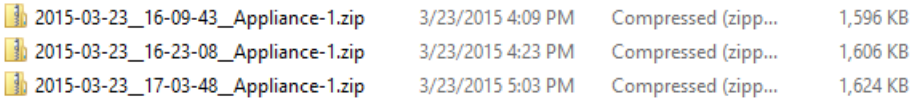
   If the server is not chosen as governor it will just jump to the cleanup phase for the log files and then exit. Within the log file this is reflected in within the following lines:

   ```
   2015-03-23__16-03-53  |  ERROR      |  elect-governor : this server was not elected. Function returns >false<.
   2015-03-23__16-03-53  |  WARNING    |  stage-2 : exiting script because I was not the governor.
   2015-03-23__16-03-53  |  VERBOSE    |  stage-2 : setting all stages to completed=true.
   ```

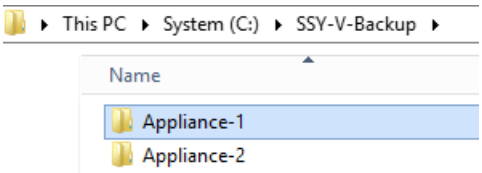
3. If the script remains running it creates a backup of the currently defined backup-path
4. Script sets the backup path to a temporary backup location. This looks like this:

   Please note that the hostname will be added as parent directory within the provided backup path.
5. Script executes the configuration-backup which will store the configuration of each node in its local folder.
6. Script will reset the original backup-configuration. If the configuration was empty before it will set the path to "local backup path\hostname". **Important**: The GUI does not immediately reflect these changes. So do not be confused if the temporary path is still shown in the GUI. Latest if the GUI is closed and reopened the changes can be seen.
7. Script will export the DCSObjectmodel.xml and copy to all the remote servers
8. Script will compress the temporary folder to a ZIP-file in the parent folder with "time-stamp__hostname.zip":

   If successful the temporary folder will be deleted
9. Script will now sync the configuration-backup to all the servers in the server group. On each server within the provided backup-path several folders should show up:

10. If a remote backup path is provided the backups also get synced to that share.

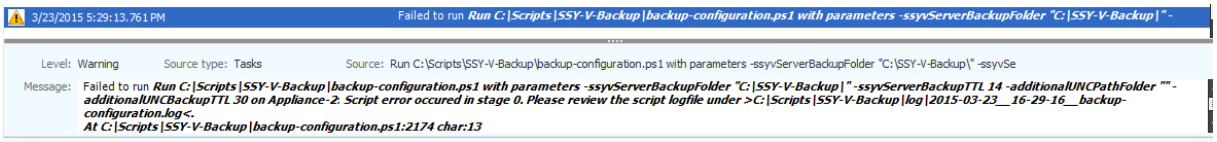
11. After that the cleanup process starts
    a. First for all the backups in the server group (TTL local)
    b. If remote backup is used also clean up there (TTL remote)
    c. Log file cleanup

FAQ

Q: If I run the script manually it works but if I use SSY scheduler it fails constantly. What´s wrong?
A: Note that SSY scheduler uses DCSadmin account which may not have sufficient privileges on the remote server / share. Check permissions first before retrying.

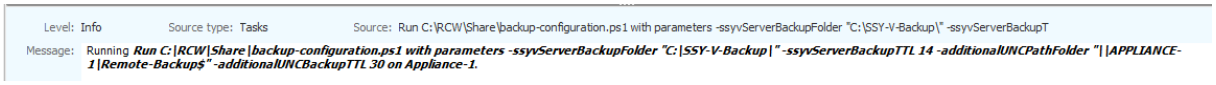
Q: I have connected remote-server group for asynchronous replication. How does that work?
A: Remote nodes are filtered through the script-run. You need to implement a separate script / backup instance for the remote node.

Q: How can I check if the script failed?
A: The script will push an error into the SSY log and creates an alert if something goes wrong. This will look like this:



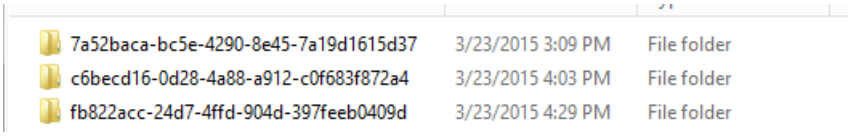
The message contains a hint where the log file is stored for that run.

If the script ran fine you will just receive informational records like this:



Q: My backups are not getting created.
A: This may happen if script run fails on the server that is currently the governor. Check the log file provided in the message for further information what is going wrong.

Q: I am having a maintenance window and my backup fails.
A: This is expected. Due to the sync mechanism not being complete the script will produce warnings / errors during that time frame. As soon as all servers in the server group are available again this behavior will disappear. You may also find incomplete backups like shown below:



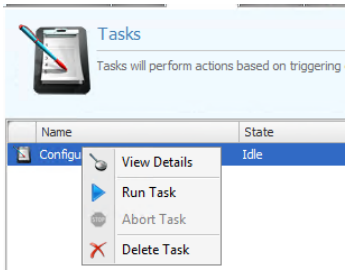
As other (successful) backups they get deleted once their TTL is reached.

Q: Script says that the backup location has been reset but it is not reflected in the GUI.
A: This is expected. You need to force a refresh by closing the GUI. Anyways this can be ignored.

Q: Script constantly fails in "stage 1" and complains about SHA checksum error. What is that all about?
A: Someone modified the script (either on purpose or by accident) and the script protects itself by exiting before breaking something. Copy the script from another server over the modified one. The checksum also help with troubleshooting.

Q: I would like to manually execute a backup. What´s the best way to do it?

A: Use the SSY scheduler, right click on the task and choose “run task”:



Q: I used the backup only within server group and would like to extend to a remote backup share. How can I do this?

A: **Important:** Do this on all nodes in the server group.

Aside that it is easy. Just reissue the script in installation mode and provide the additional / updated parameters. The script will then check if the parameters are equal to the ones stored in the action of the task and recreate if necessary. Example:

```
PS C:\Scripts\SSY-V-Backup> .\backup-configuration.ps1 -installScript $true -ssy
vServerBackupTTL 100 -additionalUNCPathFolder "\\mybackupserver\ssyv" -additiona
lUNCBackupTTL 1000_
```

Output will show:

```
2015-03-23__17-54-16  |  INFORMATION  |  create-powershellscripttask : Checking if there is an action for this server >A
ppliance-1<.
2015-03-23__17-54-18  |  INFORMATION  |  create-powershellscripttask : action found but parameters do not match. Removin
g old action.
2015-03-23__17-54-18  |  SUCCESS      |  create-powershellscripttask : success.
2015-03-23__17-54-18  |  INFORMATION  |  create-powershellscripttask : creating action to perform backup script on this
server.
2015-03-23__17-54-18  |  SUCCESS      |  create-powershellscripttask : success.
```

Execution examples

These execution examples may help understanding the usage of the script:

backup-configuration.ps1

Issues the script with default parameters. The script will run a backup in the servergroup. Therefore all servers need to have an appropriate UNC-share configured. The expected name of the UNC-Share is the folder-name of the local backup path. For instance: "C:\SSY-Backup" will expect an UNC share on every SSY server with "SSY-Backup$"

backup-configuration.ps1 -overWriteLogFile $true

Will issue the script but won´t provide a time-stamp in front of the log file-name. Therefore with the next run this log file will be overwritten. This parameter will not be used if the “installscript” parameter is used.

backup-configuration.ps1 -installScript $true

Script will run the install-mode with default parameters: Backup-Folder locally to "c:\SSY-Backup". Backups are kept for 14 days.

backup-configuration.ps1 -installScript $true -additionalUNCPathFolder "\\mybackup-server\remote-share"

Script will run the install-mode with default parameters: Backup-Folder locally to "c:\SSY-Backup". Also additional copies will be stored on the UNC-path "\\mybackup-server\remote-share". Backups within the server group are kept for 14 days. On the remote-share backups are kept for 30 days.

backup-configuration.ps1 -ssyvServerBackupFolder "C:\Mybackup" -forcebackup $true

Script will run with the path of the backup "C:\Mybackup". Therefore it will expect a share with name "Mybackup$" on all other SSY servers. If the UNC path is not there on the SSY Servers in the group backup-script will fail to copy DCSobjectmodel to other SSY servers or sync backups within the server group. The script will proceed with the backup even if it is not the GOVERNOR of this servergroup. The GOVERNOR-mechanism is used that the script only runs once in the servergroup.

backup-configuration.ps1 -ssyvServerBackupTTL 44

Script will run and keep backups within the servergroup for 44 days (instead of 14 which is default). After that the backups will be deleted.

backup-configuration.ps1 -additionalUNCBackupTTL 300

Script will run and keep backups on remote backup share 300 days (instead of 30 which is default). After that the backups will be deleted.